Final Project

Incorporating Mask-Related Transmission Variability into SEIR Models

INF301: System Modeling and Simulation

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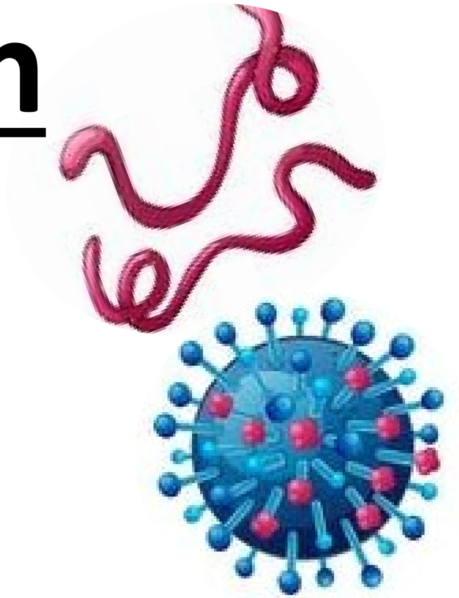
Introduction



- Viral epidemics are not uncommon in human history.
  - Severe acute respiratory syndrome coronavirus (SARS-CoV-1 virus);
  - Influenza A (H1N1 virus);
  - Middle East respiratory syndrome coronavirus (MERS-CoV virus);
  - Ebola virus disease;
  - COVID-19 (SARS-CoV-2 virus);
- These virus diseases have tested national health systems and the global governance architecture for health security.

While forecasting the spread of a virus is a significant challenge, simulating various scenarios of disease transmission within a population can be valuable for enhancing preparedness for future epidemics

Bhadoria et al., 2021, Morens et al., 2020
Wilder-Smith, 2021

Introduction

- Viral pandemics are not uncommon in human history.

In this scenario, mathematical models can be used to simulate different scenarios and, thus improving preparedness for future epidemics

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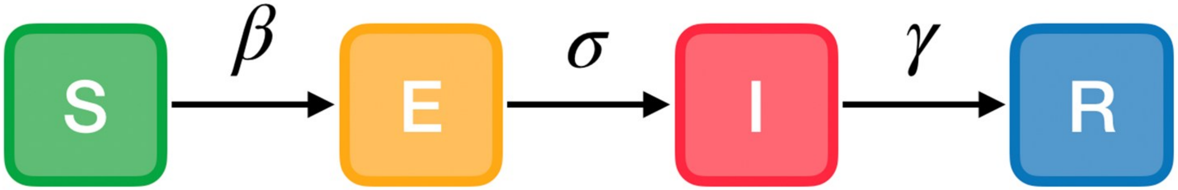
Introduction

- SEIR (Susceptible-Exposed-Infected-Recovered) model is widely used to represent the dynamics of viral epidemics in human populations.
- SEIR model divides the population into four compartments:
  - Susceptible (S);
  - Exposed (E);
  - Infected (I);
  - Recovered (R);



- Thus, transitions between compartments can be simulated using mathematical approaches, allowing hypothetical scenarios such as the impact of mask-based interventions to be evaluated.

Liu, 2023

Objective

- To model the spread of COVID-19 in São Paulo (SP), Brazil using a SEIR model;
- To incorporate, experimentally, mask-related changes in transmission rate (λ) to compare infection dynamics under different mask scenarios;
- To estimate the impact on severe cases (≈ 15% of infections);
- To assess whether mask use helps keep severe cases below the available emergency bed capacity.

SEIR Model Structure

For modeling the spread of COVID-19 transmission during the first wave (2020), four groups of individuals (compartments) were considered:

- **Compartments**:
  - **S(t)** – Susceptible,
  - **E(t)** – Exposed,
  - **I(t)** – Infectious,
  - **R(t)** – Recovered.
- **Total population**: $N = S + E + I + R$.
- **Key parameters:**
  - **λ** (infection rate),
  - **σ** (incubation rate),
  - **δ** (recovery rate).

SEIR Model Structure

- So, the system of differential equations can be written as follows:

$$\begin{cases} \dfrac{dS}{dt} & = & \dfrac{-\lambda SI}{N} \\[2em] \dfrac{dE}{dt} & = & \dfrac{-\lambda SI}{N} - \sigma E \\[2em] \dfrac{dI}{dt} & = & \sigma E - \delta I \\[2em] \dfrac{dR}{dt} & = & \delta I \end{cases}$$

Where:

S(t) – number of susceptible individuals,

E(t) – number of exposed individuals,

I(t) – number of infectious individuals,

R(t) – number of recovered individuals.

N = total of population.

λ = infection rate,

σ = incubation rate,

δ = recovery rate.

SEIR Model Structure

- Due to elevated complexity of differential equations, we can derive it to difference equations using the Euler method to numerically simulate the behavior of the system:

$$
\begin{cases}
S(t+\Delta t) &= S(t) - \frac{\lambda}{N} S(t) I(t) \Delta t \\
E(t+\Delta t) &= E(t) + \left(\frac{\lambda}{N} S(t) I(t) - \sigma E(t) \right) \Delta t \\
I(t+\Delta t) &= I(t) + \left(\sigma E(t) - \delta I(t) \right) \Delta t \\
R(t+\Delta t) &= R(t) + \delta I(t) \Delta t
\end{cases}
$$

Discrete time = step (days)

Model Parameters & Mask Scenarios

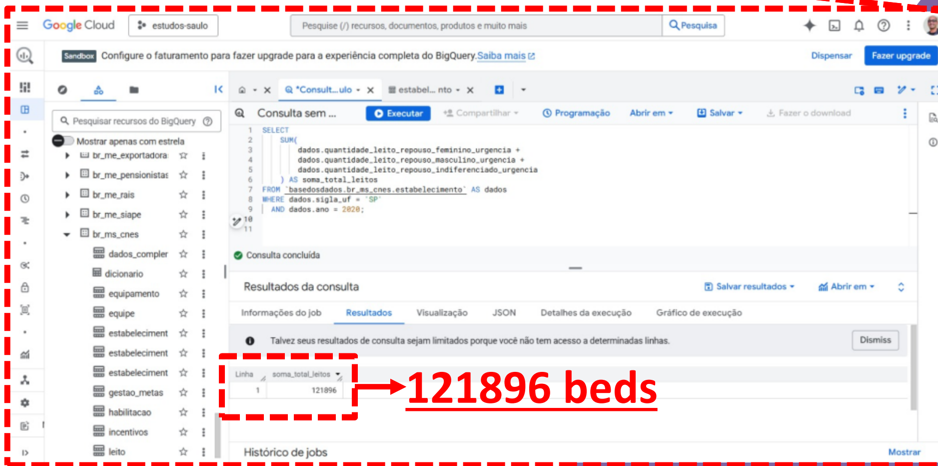
- Base epidemiological parameters (literature):

Parameter	Symbol	Value	Description	Reference
Infection Rate - No mask	$\lambda_{\text{No mask}}$	0.59	Rate at which susceptible individuals become exposed	5
Infection Rate - Cloth mask	$\lambda_{\text{Cloth mask}}$	0.42 (i.e.,-30%)	Rate at which susceptible individuals become exposed	6
Infection Rate - Cirurgical mask	$\lambda_{\text{Cirurgical mask}}$	0.30 (i.e.,-50%)	Rate at which susceptible individuals become exposed	6
Infection Rate - N95	λ_{N95}	0.18 (i.e.,-70%)	Rate at which susceptible individuals become exposed	6
Incubation Rate	σ	1/5 days	Rate at which exposed individuals become infectious	5
Recovery Rate	δ	1/10 days	Rate at which infectious individuals recover	5
Sample Size (population of SP)	N	11.9 million	Populations of SP	5

Health-care system capacity

- To obtain the number of rest/observation beds in emergency units, data from the CNES (Cadastro Nacional de Estabelecimentos de Saúde) available on basedosdados.org was accessed using a SQL query in BigQuery (Google Cloud) on the basedosdados.br_ms_cnes.estabelecimento table using the code.

```sql
SELECT
 SUM(
  dados.quantidade_leito_repouso_feminino_urgencia +
  dados.quantidade_leito_repouso_masculino_urgencia +
  dados.quantidade_leito_repouso_indiferenciado_urgencia +
 ) AS soma_total_leitos
FROM
  `basedosdados.br_ms_cnes.estabelecimento` AS dados
WHERE dados.sigla_uf = 'SP'
  AND dados.ano = '2020'
```

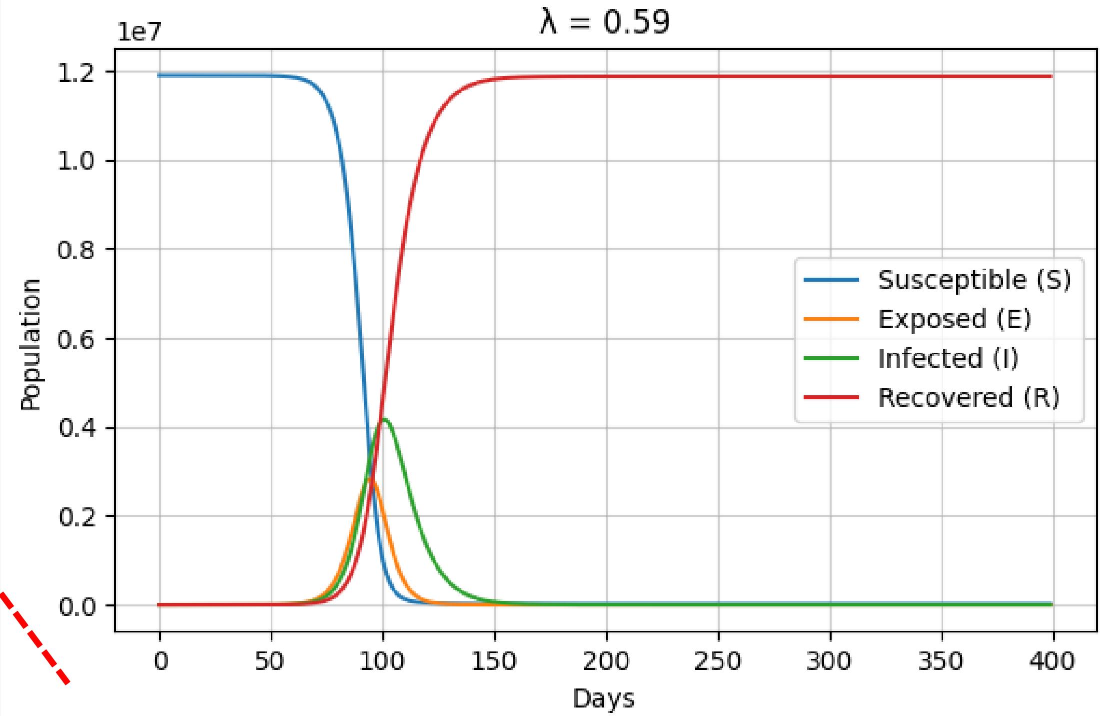


Numerical Implementation

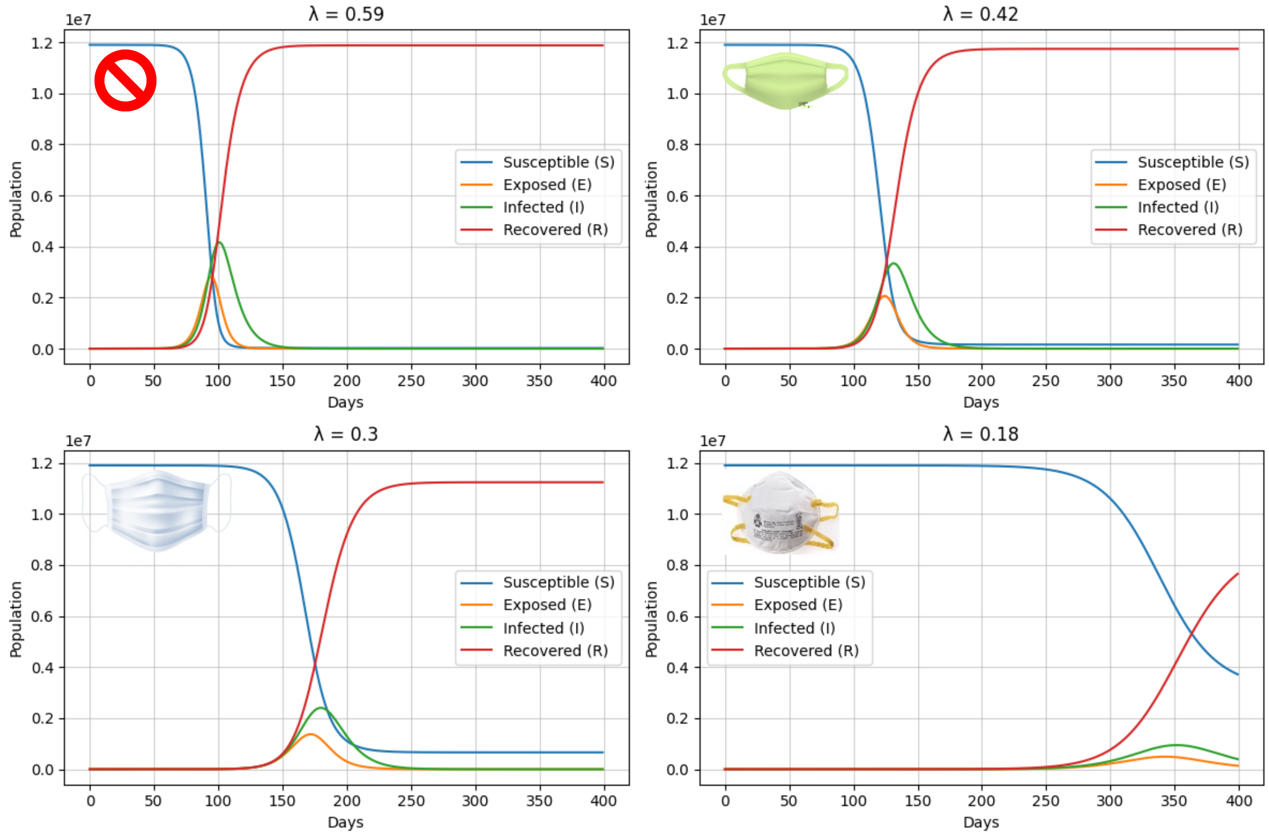
- **Python.**
  - *NumPy* for numerical computations and;
  - *Matplotlib* for visualization.
- **The model was run for a total of 400 days:**
- **Initial conditions was set up as follows:**
  - number of susceptible (i.e., zero individuals),
  - number of exposed (i.e., N - 1 individuals),
  - number of infectious (i.e., zero individuals), and
  - number of recovered individuals (i.e., zero individuals)
- **Time step of 1 day.**

Results I – SEIR Trajectories

```python
import numpy as np
import matplotlib.pyplot as plt

# Parameters
N = 1.19e7                          # total population
lambdas = [0.59, 0.42, 0.30, 0.18]  # transmission rates (no mask, cloth (-30%), surgical (-50%), N95 (-70%))
delta = 0.10                         # recovery rate (δ) ~ 10 days (1/10)
sigma = 0.20                         # incubation rate (σ) ~ 5 days (1/5)
Delta_t = 1                         # time step (days)
T = 400                             # total time (days)

# Time vector
t = np.arange(0, T, Delta_t)

# List to store all I curves
all_I = []

fig, axs = plt.subplots(2, 2, figsize=(12, 8))
axs = axs.ravel()

for idx, lambda_ in enumerate(lambdas):

    # Arrays for S, E, I, R
    S = np.zeros(len(t))
    E = np.zeros(len(t))
    I = np.zeros(len(t))
    R = np.zeros(len(t))

    # Initial conditions
    S[0] = N - 1 # First case in SP - 26 de fevereiro de 2020
    E[0] = 1
    I[0] = 0
    R[0] = 0

    # Euler iteration of discrete SEIR model
    for k in range(len(t) - 1):
        S[k+1] = S[k] - (lambda_ / N) * S[k] * I[k] * Delta_t
        E[k+1] = E[k] + ((lambda_ / N) * S[k] * I[k] - sigma * E[k]) * Delta_t
        I[k+1] = I[k] + (sigma * E[k] - delta * I[k]) * Delta_t
        R[k+1] = R[k] + delta * I[k] * Delta_t

    # Save I curve
    all_I.append(I)

    # Plot SEIR subplots
    ax = axs[idx]
    ax.plot(t, S, label="Susceptible (S)")
    ax.plot(t, E, label="Exposed (E)")
    ax.plot(t, I, label="Infected (I)")
    ax.plot(t, R, label="Recovered (R)")
    ax.set_title(f"λ = {lambda_}")
    ax.set_xlabel("Days")
    ax.set_ylabel("Population")
    ax.legend()
    ax.grid(True, alpha=0.6)

plt.tight_layout()
plt.show()
```

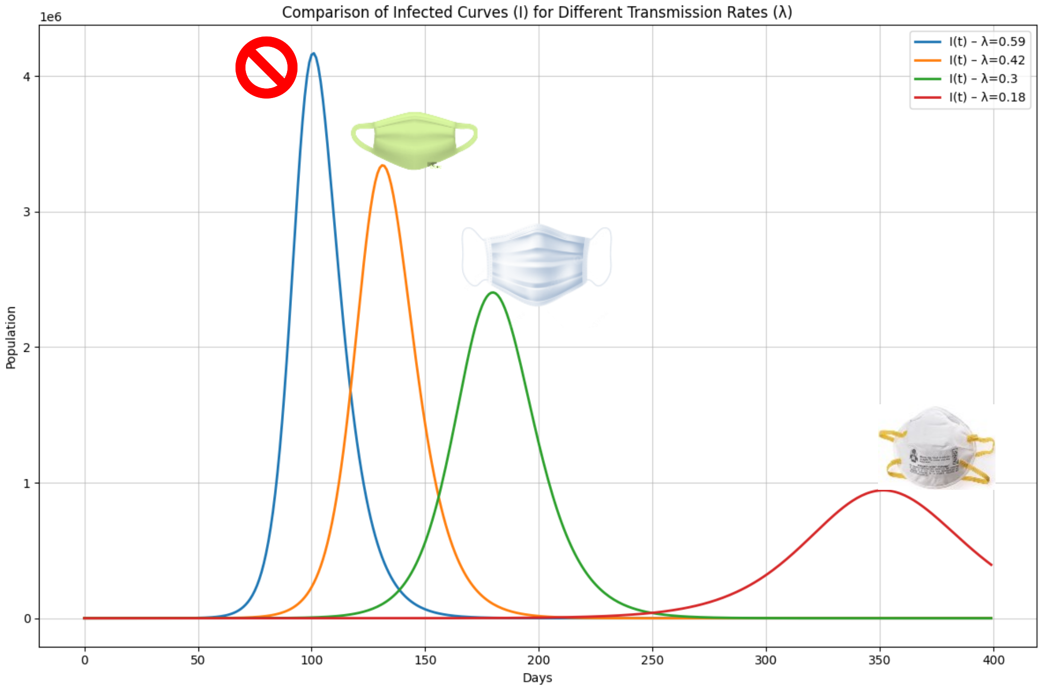


Results I – SEIR Trajectories by Mask Scenario



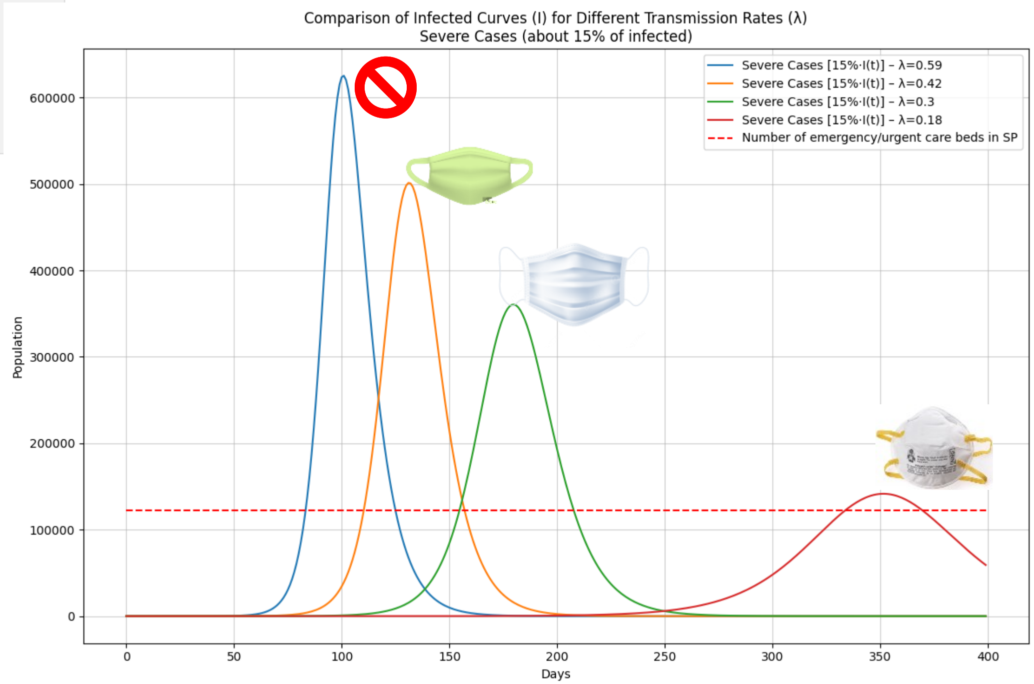
Results II – Comparison of Infected Curves I(t)

```python
# All Infected Curves
plt.figure(figsize=(12, 8))

for I_curve, lambda_ in zip(all_I, lambdas):
    # Curva total de infectados
    plt.plot(t, I_curve, label=f"I(t) - λ={lambda_}", linewidth=2)

plt.title("Comparison of Infected Curves (I) for Different Transmission Rates (λ)")
plt.xlabel("Days")
plt.ylabel("Population")
plt.grid(True, alpha=0.6)

# # Hospital bed capacity line
# plt.hlines(y=121896, xmin=0, xmax=max(t), color='r', linestyle='--',
#            label='Número de leitos de urgência/emergência em SP')

plt.legend()
plt.tight_layout()
plt.show()
```



Results III – Severe Cases and Bed Capacity

```python
# All Infected Curves
plt.figure(figsize=(12, 8))

for I_curve, lambda_ in zip(all_I, lambdas):
    # 15% dos infectados (ex.: casos que vão desenvolver sintomas graves)
    I_curve_15 = 0.15 * I_curve
    # Curva de 15% dos infectados
    plt.plot(t, I_curve_15, label=f"Severe Cases [15%·I(t)] – λ={lambda_}")

plt.title("Comparison of Infected Curves (I) for Different Transmission Rates (λ)\nSevere Cases (about 15% of infected)")
plt.xlabel("Days")
plt.ylabel("Population")
plt.grid(True, alpha=0.6)

# Hospital bed capacity line
plt.hlines(y=121896, xmin=0, xmax=max(t), color='r', linestyle='--',
           label='Number of emergency/urgent care beds in SP')

plt.legend()
plt.tight_layout()
plt.show()
```



Conclusion

- SEIR model is a useful approach to explore spread of virus diseases like COVID-19.
- As λ decreases from values representing no mask use to cloth, surgical, and finally N95/PFF2 masks, both the total number of infections and the peak prevalence of active cases are progressively reduced and delayed, demonstrating a clear flattening of the epidemic curve.
- When severe cases are approximated as 15% of infections and compared with the number of rest/observation beds available in emergency units in São Paulo, only the N95/PFF2 mask produces a peak of severe disease that approaches, rather than vastly exceeds, the existing capacity.
- Although based on a simplified model and several assumptions, results obtained across mathematical simulations reinforce the importance of high-efficacy masks as a non-pharmaceutical intervention capable of mitigating health-system overload during viral pandemics, particularly in the early phases when vaccines and specific treatments may not yet be widely available.

Limitations

Although SEIR framework is useful to explore hypothetical scenarios related to impact of mask-based interventions and to generate insights that can aid preparedness and response planning for future epidemics, some issues should be considered:

I. First, it assumes homogeneous mixing of the population, ignoring heterogeneity in contact patterns by age, occupation, household structure, or geography;

II. Second, model parameters such as transmission, incubation, and recovery rates were assumed to be constant, which may not capture rapid changes in behavior, policy measures, viral variants, or health-care capacity over time;

III. Third, classical SEIR models are deterministic and do not incorporate stochastic effects that can be relevant, especially in the early phase of an outbreak or in smaller populations.

IV. Fourth, the model usually assumes that recovered individuals acquire complete and lasting immunity, overlooking waning immunity, reinfections, and the impact of vaccination.

V. Finally, when SEIR models are used to evaluate interventions such as mask use, they often encode these effects as static reductions in the transmission rate, without explicitly accounting for imperfect adherence, incorrect use, or differences in risk across settings (e.g., households vs. workplaces)

References

1. Bhadoria P, Gupta G, Agarwal A. Viral Pandemics in the Past Two Decades: An Overview. J Family Med Prim Care. 2021;10(8):2745-50.

2. Morens DM, Fauci AS. Emerging Pandemic Diseases: How We Got to COVID-19. Cell. 2020;182(5):1077-92.

3. Wilder-Smith A. COVID-19 in comparison with other emerging viral diseases: risk of geographic spread via travel. Trop Dis Travel Med Vaccines. 2021;7(1):3.

4. Liu X. Analytical solution of l-i SEIR model-Comparison of l-i SEIR model with conventional SEIR model in simulation of epidemic curves. PLoS One. 2023;18(6):e0287196.

5. Pinto Neto O, Kennedy DM, Reis JC, Wang Y, Brizzi ACB, Zambrano GJ, de Souza JM, Pedroso W, de Mello Pedreiro RC, de Matos Brizzi B, Abinader EO, Zângaro RA. Mathematical model of COVID-19 intervention scenarios for São Paulo-Brazil. Nat Commun. 2021 Jan 18;12(1):418.

6. Floriano I, Silvinato A, Bacha HA, Barbosa AN, Tanni S, Bernardo WM. Effectiveness of wearing masks during the COVID-19 outbreak in cohort and case-control studies: a systematic review and meta-analysis. J Bras Pneumol. 2024 Jan 5;49(6):e20230003.

Thank you for your attention!

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